Lecture 10 Review: Support Vector Machines

Problem, Mathematical Modeling, and Solution

Machine Learning Tutorial

School of Engineering, Westlake University

Fall 2026

WESTLAKE UNIVERSITY | SCHOOL OF ENGINEERING

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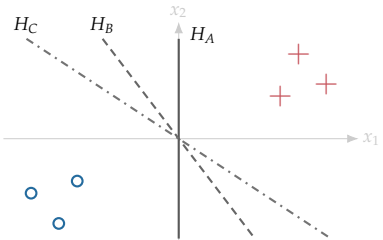
3.6. From Dual Solution to Prediction

1. Problem: Many Separating Hyperplanes

Given: $N \geq 1$ labeled inputs $x_n \in \mathbb{R}^D$, $y_n \in \{-1, +1\}$.

Learn: $w \in \mathbb{R}^D$ for the score $f_w(x) = w^\top x$ and prediction $\hat{y} = \text{sign}(f_w(x))$.

We fix $b = 0$ throughout. The decision boundary passes through the origin; $f_w(x) = 0$ is a tie.



$+: y = +1; \circ: y = -1.$

Start with the simplest case: separable data.

Each displayed hyperplane puts all positive and negative examples on their correct sides.

Many classifiers can therefore satisfy

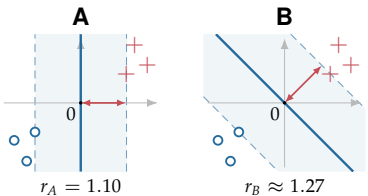
$$y_n w^\top x_n > 0 \qquad \forall n.$$

$$H_w = \{x : w^\top x = 0\}, \qquad \mathcal{S} = \{w : y_n w^\top x_n > 0 \; \forall n\}.$$

Problem: Among all separating hyperplanes, which one should we choose?

2.1. Mathematical Modeling: Geometric Objective

Among classifiers in $\mathcal{S}$, prefer a boundary whose nearest training point is farther away.



Same training inputs and two candidate separators.

First, evaluate one separator.

Its smallest point-to-boundary distance measures the clearance of its most vulnerable training point.

Then, compare separators.

Maximize that smallest distance. This protects the nearest point, rather than averaging distances.

$$r(w) = \min_{1 \le n \le N} \operatorname{dist}(x_n, H_w), \qquad \boxed{\max_{w \in \mathcal{S}} r(w).}$$

Here $r_B > r_A$, so B is preferred to A. The optimum compares *all* separators in $\mathcal{S}$.

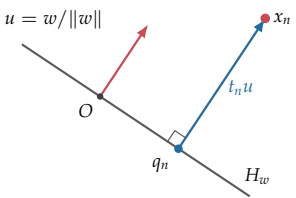
Next, derive a formula for the distances and substitute it into this same $r(w)$.

2.2. Mathematical Modeling: Explicit Objective

Same objective: $\max_{w \in \mathcal{S}} r(w)$, where $r(w) = \min_n \operatorname{dist}(x_n, H_w)$.

1. Express each distance in terms of w.

For the perpendicular foot $q_n \in H_w$, $w^\top q_n = 0$. With unit normal $u = w/\|w\|$, write $x_n = q_n + t_n u$:



$$w^\top x_n = 0 + t_n \frac{w^\top w}{\|w\|} = t_n \|w\|,$$

$$\operatorname{dist}(x_n, H_w) = |t_n| = \frac{|w^\top x_n|}{\|w\|}.$$

2. Substitute into $r(w)$. Since $w \in \mathcal{S}$, $m_n = y_n w^\top x_n > 0$ gives $|w^\top x_n| = m_n$. The norm is positive and independent of n:

$$r(w) = \min_n \frac{|w^\top x_n|}{\|w\|} = \min_n \frac{y_n w^\top x_n}{\|w\|} = \frac{\min_n y_n w^\top x_n}{\|w\|}.$$

$$\Longrightarrow \quad \boxed{\max_{w \in \mathcal{S}} r(w) = \max_{w \in \mathcal{S}} \frac{\min_n y_n w^\top x_n}{\|w\|}.}$$

2.3. Mathematical Modeling: Fixing the Scale

Previous objective:

$$\max_{w \in \mathcal{S}} r(w) = \max_{w \in \mathcal{S}} \frac{\min_n y_n w^\top x_n}{\|w\|}.$$

1. Positive rescaling does not change the objective. For any $c > 0$,

$$r(cw) = \frac{\min_n y_n (cw)^\top x_n}{\|cw\|} = \frac{c \min_n y_n w^\top x_n}{c\|w\|} = r(w).$$

The scores change, but $H_{cw} = H_w$: these are different scales for the same separator.

2. Normalize the numerator. For any $w \in \mathcal{S}$, let $\gamma = \min_n y_n w^\top x_n > 0$.

$$\widetilde{w} = \frac{w}{\gamma} \quad \Longrightarrow \quad \min_n y_n \widetilde{w}^\top x_n = \frac{\gamma}{\gamma} = 1, \qquad r(\widetilde{w}) = r(w).$$

Every separating boundary has this normalized representative; none is lost.

Writing w for normalized $\widetilde{w}$, the same geometric problem becomes

$$\max_{w} \frac{1}{\|w\|} \qquad \text{s.t.} \qquad \min_n y_n w^\top x_n = 1.$$

2.4. Mathematical Modeling: Hard-Margin SVM

Keep $\min_n m_n(w) = 1$ from the previous page, with $m_n(w) = y_n w^\top x_n$:

$$\max_w \frac{1}{\|w\|} \iff \min_w \|w\| \iff \min_w \frac{1}{2}\|w\|^2, \qquad \text{same optimal } w.$$

Enlarge the feasible set: replace $\min_n m_n(w) = 1$ by $m_n(w) \geq 1$ for all n. Let $\bar{w}$ be an **optimal solution of this enlarged problem**, and set $\gamma = \min_n m_n(\bar{w}) \geq 1$.

Case 1: $\gamma = 1$**.** Then $\bar{w}$ satisfies the original equality constraint. It minimizes the objective over the larger set, so it also minimizes it over the original subset.

Case 2: $\gamma > 1$**.** Construct $w' = \bar{w}/\gamma$. Then

$$\min_n m_n(w') = 1, \qquad \frac{1}{2}\|w'\|^2 = \frac{\frac{1}{2}\|\bar{w}\|^2}{\gamma^2} < \frac{1}{2}\|\bar{w}\|^2.$$

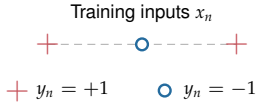
But w' is also enlarged-feasible and has a lower objective: **contradiction to optimality of** $\bar{w}$**.**

Thus only Case 1 is possible, and the two minimum values are equal. Every original optimum is also enlarged-feasible with that same value, so **the optimal solutions coincide**.

$$\boxed{\min_w \frac{1}{2}\|w\|^2 \qquad \text{s.t.} \qquad y_n w^\top x_n \geq 1 \quad \forall n.}$$

2.5. Mathematical Modeling: Soft-Margin SVM

1. When hard-margin fails: nonseparable data.



The negative input lies between two positive inputs. **No hyperplane separates all three.**

So $y_n w^\top x_n \geq 1$ cannot hold for every n.

2. Solution: allow margin shortfalls. Introduce one slack $\xi_n \geq 0$ per input:

$$\underbrace{y_n w^\top x_n \geq 1}_{\text{hard requirement}} \quad \longrightarrow \quad \underbrace{y_n w^\top x_n \geq 1 - \xi_n, \quad \xi_n \geq 0}_{\text{allow a margin shortfall}}.$$

$\xi_n = 0$ keeps the hard requirement; $\xi_n > 0$ permits a shortfall in **score units**.

3. New optimization problem. Keep the norm penalty and add a cost for slack:

$$\min_{w,\xi} \quad \underbrace{\frac{\lambda}{2}\|w\|^2}_{\text{norm penalty}} + \underbrace{\frac{1}{N}\sum_{n=1}^{N} \xi_n}_{\text{average slack cost}}$$

$$\text{s.t.} \quad y_n w^\top x_n \geq 1 - \xi_n, \qquad \xi_n \geq 0 \quad \forall n.$$

$\lambda > 0$ balances the two costs. **Next:** choose the best slacks for a fixed w.

2.6. Mathematical Modeling: Primal Objective

1. Recall page 8: optimize w and ξ together.

$$\boxed{\begin{aligned} \min_{w,\xi} \quad & \frac{\lambda}{2}\|w\|^2 + \frac{1}{N}\sum\nolimits_{n=1}^{N} \xi_n \\ \text{s.t.} \quad & y_n w^\top x_n \geq 1 - \xi_n, \quad \xi_n \geq 0 \quad \forall n. \end{aligned}}$$

2. Same feasible pairs: fix w, minimize over ξ, then optimize w.

$$= \min_{w} \left[\frac{\lambda}{2}\|w\|^2 + \min_{\substack{\xi:\ \xi_n \geq 0 \\ \xi_n \geq 1 - y_n w^\top x_n,\ \forall n}} \frac{1}{N}\sum\nolimits_{n=1}^{N} \xi_n \right].$$

The inner problem has independent constraints and positive coefficients $1/N$, so

$$\xi_n^*(w) = \max\{0, 1 - y_n w^\top x_n\} = [1 - y_n w^\top x_n]_+.$$

3. Substitute $\xi^*(w)$: the hinge form of the same problem.

$$= \min_{w} \underbrace{\left[\frac{\lambda}{2}\|w\|^2 + \frac{1}{N}\sum\nolimits_{n=1}^{N} [1 - y_n w^\top x_n]_+ \right]}_{L(w)}.$$

Same optimal value and optimal w; recover the slacks as $\xi_n^*(w^*)$.

3.1. Mathematical Solution: Hinge Reformulation

To expose a quadratic problem in w, rewrite each hinge term. For any real z, maximize αz over $0 \leq \alpha \leq 1$:

Value of z	Maximizing α	$\max_{0\leq\alpha\leq 1} \alpha z$
$z > 0$	1	z
$z < 0$	0	0
$z = 0$	any $\alpha \in [0, 1]$	0

$$\Longrightarrow \qquad \boxed{[z]_+ = \max_{0\leq\alpha\leq 1} \alpha z.}$$

Apply the identity with $z_n = 1 - y_n w^\top x_n$:

$$\begin{aligned}\frac{1}{N}\sum_n [z_n]_+ &= \frac{1}{N}\sum_n \max_{0\leq\alpha_n\leq 1} \alpha_n z_n \\ &= \max_{\alpha\in[0,1]^N} \frac{1}{N}\sum_n \alpha_n z_n.\end{aligned}$$

The last equality holds because every α_n can be chosen independently.

The model is unchanged. ξ_n measured an allowed shortfall; α_n re-expresses the hinge term and will become a sample weight.

3.2. Mathematical Solution: Min–Max Form

The norm term is independent of α. Define

$$G(w,\alpha) = \frac{\lambda}{2}\|w\|^2 + \frac{1}{N}\sum_n \alpha_n(1 - y_n w^\top x_n), \qquad \alpha \in [0,1]^N.$$

$$L(w) = \max_{\alpha\in[0,1]^N} G(w,\alpha), \qquad \boxed{p^* = \min_w \max_{\alpha\in[0,1]^N} G(w,\alpha).}$$

For each w, the inner maximum computes its exact loss; the outer minimum selects w.

Why seek a dual? For fixed α,

$$G(w,\alpha) = \underbrace{\frac{\lambda}{2}\|w\|^2}_{\text{quadratic in } w} \underbrace{- \left(\frac{1}{N}\sum_n \alpha_n y_n x_n\right)^\top w}_{\text{linear in } w} + \underbrace{\frac{1}{N}\sum_n \alpha_n}_{\text{constant in } w}.$$

We can minimize this quadratic analytically. This motivates the **dual**:

$$\boxed{g(\alpha) = \min_w G(w,\alpha), \qquad d^* = \max_{\alpha\in[0,1]^N} g(\alpha).}$$

The optimization order has changed. We must establish whether $d^* = p^*$.

3.3. Mathematical Solution: Slater's Theorem

For every w and feasible α, minima and maxima give

$$\underbrace{\min_{v} G(v,\alpha)}_{g(\alpha)} \le G(w,\alpha) \le \underbrace{\max_{\beta\in[0,1]^N} G(w,\beta)}_{L(w)} \quad\Longrightarrow\quad d^* \le p^*.$$

Theorem 1 (Slater; soft-margin SVM form)

For $\lambda > 0$, consider the convex primal problem with optimal value

$$p^* := \inf_{\substack{w\in\mathbb{R}^D,\ \xi\in\mathbb{R}^N \\ y_n w^\top x_n \ge 1-\xi_n,\ \xi_n\ge 0\ (\forall n)}} \left[\frac{\lambda}{2}\|w\|^2 + \frac{1}{N}\sum_{n=1}^{N}\xi_n\right].$$

Let $g(\alpha) := \inf_w G(w,\alpha)$ and $d^* := \sup_{\alpha\in[0,1]^N} g(\alpha)$.

*Suppose $p^* \in \mathbb{R}$ and the **Slater condition** holds:*

$$\exists(\bar{w},\bar{\xi}):\quad y_n\bar{w}^\top x_n > 1-\bar{\xi}_n,\quad \bar{\xi}_n > 0,\qquad \forall n.$$

*Then **strong duality** holds and the **dual optimum is attained**:*

$$\boxed{d^* = p^*,\qquad \exists\alpha^* \in [0,1]^N \text{ such that } g(\alpha^*) = p^*.}$$

3.4. Mathematical Solution: Checking the SVM

Check the conditions of the theorem for our soft-margin problem:

$$\begin{aligned} \min_{w,\xi} \quad & \frac{\lambda}{2}\|w\|^2 + \frac{1}{N}\sum\nolimits_{n=1}^{N} \xi_n \\ \text{s.t.} \quad & y_n w^\top x_n \geq 1 - \xi_n, \quad \xi_n \geq 0 \quad \forall n. \end{aligned}$$

1. Convex primal. For $\lambda > 0$, the objective is a convex quadratic plus a linear term; all constraints are affine in (w, ξ).

2. Finite optimal value. Feasible costs are nonnegative; $w = 0, \xi_n = 2$ is feasible and costs 2. Therefore $0 \leq p^* \leq 2$.

3. Slater condition. Use the same point $\bar{w} = 0$, $\bar{\xi}_n = 2$ for every n:

$$y_n \bar{w}^\top x_n = 0 > 1 - 2 = -1, \qquad \bar{\xi}_n = 2 > 0 \quad \forall n.$$

Every inequality is strict, exactly as required in the theorem.

All conditions hold. By Slater's theorem, we have strong duality:

$$\boxed{d^* = p^*, \qquad \exists \alpha^* \in [0,1]^N : \; g(\alpha^*) = p^*.}$$

The dual reaches the original primal optimum; we can proceed to solve it.

3.5. Mathematical Solution: Deriving the Dual

1. Fix $\alpha \in [0,1]^N$. Minimize the same G over w:

$$G(w,\alpha) = \frac{\lambda}{2}\|w\|^2 + \frac{1}{N}\sum_{n=1}^{N} \alpha_n(1 - y_n w^\top x_n).$$

$$\nabla_w G = \lambda w - \frac{1}{N}\sum_n \alpha_n y_n x_n = 0 \quad \Longrightarrow \quad \boxed{w(\alpha) = \frac{1}{\lambda N}\sum_n \alpha_n y_n x_n.}$$

Because $\lambda > 0$, G is strictly convex in w: this is its unique minimum.

2. Substitute back into G. Use $\frac{1}{N}\sum_n \alpha_n y_n x_n = \lambda w(\alpha)$:

$$g(\alpha) = G(w(\alpha),\alpha) = \frac{\lambda}{2}\|w(\alpha)\|^2 + \frac{1}{N}\sum_n \alpha_n - \left(\frac{1}{N}\sum_n \alpha_n y_n x_n\right)^\top w(\alpha)$$

$$= \frac{1}{N}\sum_n \alpha_n - \frac{\lambda}{2}\|w(\alpha)\|^2$$

$$= \boxed{\frac{1}{N}\sum_n \alpha_n - \frac{1}{2\lambda N^2}\left\|\sum_n \alpha_n y_n x_n\right\|^2.}$$

3. Optimize only α: $\boxed{d^* = \max_{\alpha\in[0,1]^N} g(\alpha) = p^*.}$ (concave quadratic).

3.6. Mathematical Solution: From Dual Solution to Prediction

From dual solution to prediction: $\alpha^* \longrightarrow w^* \longrightarrow \hat{y}$.

1. Solve the previous page's dual for the optimal sample weights:

$$\boxed{\alpha^* \in \arg\max_{\alpha \in [0,1]^N} g(\alpha).}$$

2. Recover the primal classifier.

For justification, let w^* be a primal optimum. Strong duality and the definitions of g, L give

$$p^* = g(\alpha^*) \leq G(w^*, \alpha^*) \leq L(w^*) = p^*.$$

All values are equal, so w^* minimizes $G(w, \alpha^*)$ over w.
The previous page gives this unique minimizer:

$$\boxed{w^* = w(\alpha^*) = \frac{1}{\lambda N} \sum\nolimits_{n=1}^{N} \alpha_n^* y_n x_n.}$$

3. Use the recovered classifier on a new input x:

$$\boxed{\hat{y} = \operatorname{sign}\left(w^{*\top} x\right).}$$

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